

Talented and dedicated team

Centre for Sustainable Mobility

We are a group of people who work with

data and make the most out of it.

Targeting the central problems of

traffic, breakdowns and bus

availability, we utilize data that

solves one of the most important

problems for major cities. Deriving

data, applying our algorithms and

representing it in the most effective

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manner is our goal! Delhi has been suffering from serious

problems related

to traffic congestion, pollution as well as road

accidents. There is a need for both serious researches

on these issues as well as work on the developmental

aspects involving closer collaboration among the

Government, academia and industry. With this objective,

IIT-Delhi, in collaboration with Delhi Knowledge

Development Foundation (DKDF) is establishing a 'Centre

of Sustainable Mobility' as a Centre of Excellence at

the Institute. The new research centre will focus on solving the

problem of urban transportation in Delhi. The problem of

sustainable mobility is a complex problem that requires

design and optimisation, relying on data analytics and interdisciplinary research that includes computer science, artificial intelligence, electronics engineering as well as design. IIIT-Delhi has faculty members specialising on various research areas across these disciplines. The institute provides an adequate infrastructure with world class facilities to aid such advanced research and learning. The researchers at IIIT-Delhi have been working with the

Transport department of the GNCTD as well as various transit agencies like DTC, DIMTS and Delhi Metro over the last 3 years to create systems like Open Transit Data, Open Ticketing Framework for contactless ticketing in buses, depot management tool for the DTC etc. The faculty of the institute also worked with the World Bank for Chennai transit agencies as well southern railways to collect, clean and open transit data. The Centre of Sustainable Mobility, through its research

focus, will not only enhance public transit, but will also save lives by preventing road accidents. Road accidents kill more than 1.5 million people every year in India and affect more than 1 crore people with serious injuries every year. The loss of GDP due to road accidents is estimated to be around 2% only. Many of these unfortunate outcomes can be reduced by creating a better transport infrastructure in the county and the

Centre of Sustainable Mobility will strive towards it.

Moreover, sustainable public transit can also play a key role in decreasing the pollution in Delhi by removing cars from the roads and enabling people to avail public transit. The centre will also open up new avenues and courses for training students in the field of public transportation, electric vehicles as well as road safety. There is a greater need for the new age courses to be connected to the SDG goals of the UN. Dr. Pravesh Biyani, the Faculty in charge of the

Centre

on Sustainable Mobility has said “[We have been engaging with DTC, DIMTS and the transport department for 3 years. We have successfully executed the country's biggest real-time open data project as well as contactless ticketing project that is now rolled out in all the buses of Delhi. We hope to take our engagement to the next level through this centre. We also hope that technologies we build here are used not only in India but throughout the world. The main goal of this centre

is to promote public transit as much as we can". During the announcement of the new Research Centre,

Director, Prof. Ranjan Bose said, “From the perspective of mobility, we are gradually moving towards an increasingly connected, autonomous, electrified, and shared future. The Centre will catalyze research, development and entrepreneurial activities in the area

of smart/ sustainable mobility, with a focus on mobility-related problems of Delhi/ NCR. " We are a group of people who work with data and

make the most out of it. Targeting the central problems of traffic, breakdowns and bus availability, we utilize data that solves one of the most important problems for major cities. Deriving data, applying our algorithms and representing it in the most effective manner is our goal! The Centre of Sustainable Mobility, through its research focus, will not only enhance public transit, but will also save lives by preventing road accidents. Road accidents kill more than 1.5 million people every year in India and affect more than 1 crore people with serious injuries every year. The loss of GDP due to road accidents is estimated to be around 2% only. Many of these unfortunate outcomes can be reduced by creating a better transport infrastructure in the country and the Centre of Sustainable Mobility will strive towards it. Moreover, sustainable public transit can also play a key role in decreasing the pollution in Delhi by removing cars from the roads and enabling people to avail public transit. The centre will also open up

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20

YEARS OF EXPERIENCE

98

people who work with Us

56

courses for training

990

CUSTOMER WORLDWIDE

Fight Against Coronavirus

Fight Against Coronavirus

Objective

To develop and maintain the IT platforms

(website and app) for COVID-19
management in Delhi.

Description

In a bid to counter the COVID-19

pandemic and to make the dissemination
of vital information easier, the
government of National Capital Territory

(NCT) of Delhi and the Health and Family Welfare Department has constituted a multi-disciplinary Technology Innovation Group. This group will develop and maintain the IT platforms (website and app) for COVID-19 management in Delhi. For this purpose, the government has sought help from three Indraprastha Institute of Information Technology (IIIT), Delhi professors (Dr. Pravesh Biyani Associate Professor, IIIT Delhi, Dr. Tavpritesh Sethi Assistant Professor, IIIT Delhi, and Prof. Ponnurangam Kumaraguru, Professor, IIIT Delhi) to develop and maintain the IT infrastructure necessary for disseminating information regarding COVID-19. Technology innovation has a significant role to play in the fight against COVID-19, in strengthening the country's foundations, and in predicting and combating the spread of future infectious diseases. With the ongoing global health crisis, science has been at the forefront of the battle against COVID-19 and our nation is fortunate to

be making great strides in policy and

technology to tackle this pandemic. CollaborationsDelhi

Government

Data Trust

Data Trust

Objective

To create a platform with mobility data

from various transport providers.

Description

The mobility data trust is an effort to

make India's mobility data from

government agencies, private agencies

and volunteering citizens available to

developers and analysts across the

nation so we can fast-forward the

development of digital solutions to our

country's transport woes.

Chartr Android App

Chartr Android App

Objective

An app which provide directions using

static / realtime feed of buses, bus

route and metro line details and provide the eta of buses in realtime.

Description

There already exist many apps which

provide directions but they use only static schedules. We on the other hand provide directions not only using static schedule but also realtime bus feed. Our app provide realtime location of buses running of different routes and their eta on bus stops.

Key Features

Last Mile Connectivity

Last Mile Connectivity

Objective

Collecting information about Last Mile

Connectivity option available in Delhi.

Description

Almost every person uses last mile

connectivity options to go from a place to their office/home/work place. But surprisingly there is no data available for such mode anywhere. There are

different options available like

E-Richshaw, cabs, autos, etc. We are working towards

collecting such

data available through out Delhi. This

data will be helpful in better planning

journey and improving transport

condition. Current StatusOngoing

Open Data policy for transport in Tamil Nadu

Open Data policy for transport in Tamil

Nadu

Objective

To adopt a policy for open data sharing

for Public Transport operations in Tamil

Nadu and open MTC & CMRL fleet location

data for all vehicles.

Description

Open Data underpins the technology

advantages of transport. Open Data is

what allows users to understand the

transport services better, connect to

services and riders, allow for easy

payments, monitor transit performance

and encourage more multimodal

accessibility. Open data will allow the

open

GoTN to leverage an international ecosystem of innovation in transport by leveraging planning and citizen engagement tools and foster a new era of innovation in transport services. As a first step to establish this

data ecosystem, the Ministry of Transport, Government of Tamil Nadu intends to utilize the OGD portal to provide public access to static and real-time data relating to public transportation that will help (a) improve passenger experience through the development of mobile applications providing real-time ETA for buses, trains and multi-modal journey planning services, (b) inputs from urban and transport planners from analyzing such open data, that can result in more efficient design of services. For the data to be provided in consistent, reliable, usable and scalable format to successfully meet the need from varied use cases, it is essential to adopt certain standards and processes for creation, maintenance and sharing of the data. In order to achieve

the same, the Ministry of Transport wishes to engage a consulting firm to recommend policy actions and steps to share public transport data on the OGD platform. This engagement will be supported by the World Bank as part of the proposed Chennai City Partnership.

DST Air Quality Monitoring

DST Air Quality Monitoring

Objective

To create a system of vehicle-mounted sensors that performs accurate air quality sensing.

Description

We propose the creation of a dense air quality map by performing a spatiotemporal sampling using only a few moving sensors followed by the suggested matrix completion technique. This is possible as the underlying low rank and slowly time-varying structure of the air quality data can be leveraged to create models that facilitate an effective Spatio-temporal extrapolation. We claim

that to obtain a dense air quality map, the cities may require only a small fraction of moving sensors as compared to static sensors. Therefore moving air quality measurement could lead to an effective strategy against air pollution and climate change.

Results

To identify the source of local

pollution and its correlation with key geographical, meteorological, and local parameters (assuming pollution inventory). To create a web portal for effective dissemination, visualization and analysis for the collected and extrapolated data.

Key Features

Lorem Ipsum is simply dummy text of the

printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining

essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

MyBus

MyBus

Objective

A web application displaying the real-time location of buses in Delhi with filters based on routes and registration numbers.

Description

The dashboard displays the real-time location of all cluster buses in Delhi with options to filter buses based on their routes and registration number

Odd Even Analysis

Odd Even Analysis

Objective

The study presents a quantitative

analysis of the odd-even road rationing
policy on traffic congestion in Delhi.

Description

The study presents a quantitative

analysis of the odd-even road rationing
policy on traffic congestion in Delhi.

The policy was implemented from 4th
November 2019 to 15th November 2019. The
study considers three phases viz: (a)
PreDiwali, (b) Diwali, (c) Odd-Even.

Using Delhi's Open TransitData platform,
we compare the travel speeds and travel
times for the three phases.

Results

It is found that there was an increase

of 15-20% in travel speeds during the
odd-even phase in comparison to regular
days. The travel times decreased by a
similar percentage. In general, travel
times were comparable to private

vehicles on regular days. We are a group of people who work
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To each route its own ETA: A generative modeling framework for ETA prediction.

Authors:-Charul Paliwal,

Pravesh Biyani Conference/Journal:-ITSC 2019

Weighted-A* Based Energy Efficient Resource Allocation in G.Fast.

Authors:-Ankita Raj, Pravesh

Biyani Conference/Journal:-IEEE

Trans. Communications 67(11): 7744-7754 (2019)

Low Complexity Training Methods for Common Mode

Aided Cancellation of Intermittent Alien Noise in Downstream VDSL.

Authors:-Ramanjit Ahuja,

Pravesh Biyani Conference/Journal:-IEEE

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Benchmark Dataset for Timetable Optimization of

Bus Routes in the City of New Delhi.

Authors:-Gain home equity by

paying sit amet seeturlit sed dolor se equity

Conference/Journal:-CoRR

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Improved Real Time Ride Sharing via Graph

Coloring.

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Online Variational Bayesian Subspace Filtering.

Authors:-Online Variational

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2019

Delhi: Track your bus for free as GPS

feeds go live

IIIT-Delhi, Cornell University test new

transport management system

Delhi Metro's Magenta Line set to ease

connectivity but holistic solution

needed to tackle congestion woes

Data sharing is at the heart of

commuter centric cities

Govt to begin e-ticketing system for

its buses to minimise contact

Contact-less: Pay through an app for

your bus ticket soon

Soon, pay for bus ticket in Delhi via a

QR code on your phone

Public buses moved faster during

odd-even, says study

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